

CAMBRIDGE HARBOR, MD, USA

WMO: 997297

Lat: 38.5739N

Lon: 76.0686W

Elev: 20

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.9	21.3	N/A	N/A	N/A	N/A	N/A	N/A	28.8	27.3	26.4	27.5	8.9	310	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	91.0	N/A	88.8	N/A	86.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	7.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.9	22.0	18.9	14.4	95.0	4.9	2.2	10.8	96.6	8.0	97.9	5.2	99.2	1.6	100.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.5	36.3	38.1	44.9	57.5	66.6	75.3	79.7	77.7	71.4	61.1	49.7	42.6
	DBStd	16.75	9.09	8.86	8.45	7.79	6.89	5.44	4.17	3.97	5.43	7.27	7.36	8.14
	HDD50	1363	434	347	205	19	0	0	0	0	0	5	94	259
	HDD65	3907	890	753	623	248	65	3	0	0	8	162	461	694
	CDD50	4477	9	14	48	244	516	760	920	859	643	351	83	30
	CDD65	1546	0	0	1	23	116	313	455	394	201	43	0	0
	CDH74	11959	0	1	8	157	628	2361	4398	3160	1100	146	0	0
	CDH80	3574	0	0	0	34	124	693	1556	912	235	20	0	0
Wind	WSAvg	7.6	7.2	7.4	8.5	8.5	7.2	6.9	6.1	6.5	7.7	8.5	8.8	8.4
Precipitation	PrecAvg	46.6	3.6	3.2	4.1	3.5	4.4	3.6	4.9	4.7	3.8	3.3	3.5	3.6
	PrecMax	66.9	7.6	7.8	8.0	7.1	9.8	11.7	12.4	10.7	10.4	8.0	8.5	8.9
	PrecMin	31.7	0.4	0.6	1.8	0.6	0.8	0.6	1.4	0.5	0.6	0.0	0.5	0.6
	PrecStd	8.7	1.5	1.8	1.7	1.5	2.4	2.4	2.5	3.1	2.5	1.9	2.1	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.6	66.9	74.6	84.3	87.4	92.9	95.5	93.0	89.1	82.5	70.8	68.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	59.2	61.2	67.1	78.6	83.7	89.0	91.8	89.4	85.9	77.8	67.4	63.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	54.8	56.4	62.2	73.8	80.4	86.6	89.3	87.2	82.9	74.9	64.2	59.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	49.6	50.8	57.4	69.8	77.5	84.0	87.3	85.0	80.1	71.9	61.0	55.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	12.4	12.6	13.5	15.2	14.3	12.9	12.5	11.9	12.1	12.3	12.1	11.2
		MCDBR	21.0	20.8	21.3	20.5	16.6	15.7	14.6	14.1	14.7	15.1	15.9	16.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.313	0.320	0.356	0.396	0.441	0.489	0.516	0.508	0.413	0.360	0.327	0.312
	taud		2.485	2.459	2.406	2.312	2.219	2.132	2.069	2.077	2.319	2.469	2.521	2.528
	Ebn at Noon		275	288	286	280	268	254	246	244	264	270	268	267
	Edh at Noon		25	30	34	40	45	49	52	50	37	29	24	23
All-Sky Solar Radiation	RadAvg		687	952	1270	1611	1828	2009	1956	1748	1419	1070	776	604
	RadStd		55	100	106	116	208	152	94	103	147	114	68	56

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (85 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	+151

Nomenclature: See separate page